

## Task -1

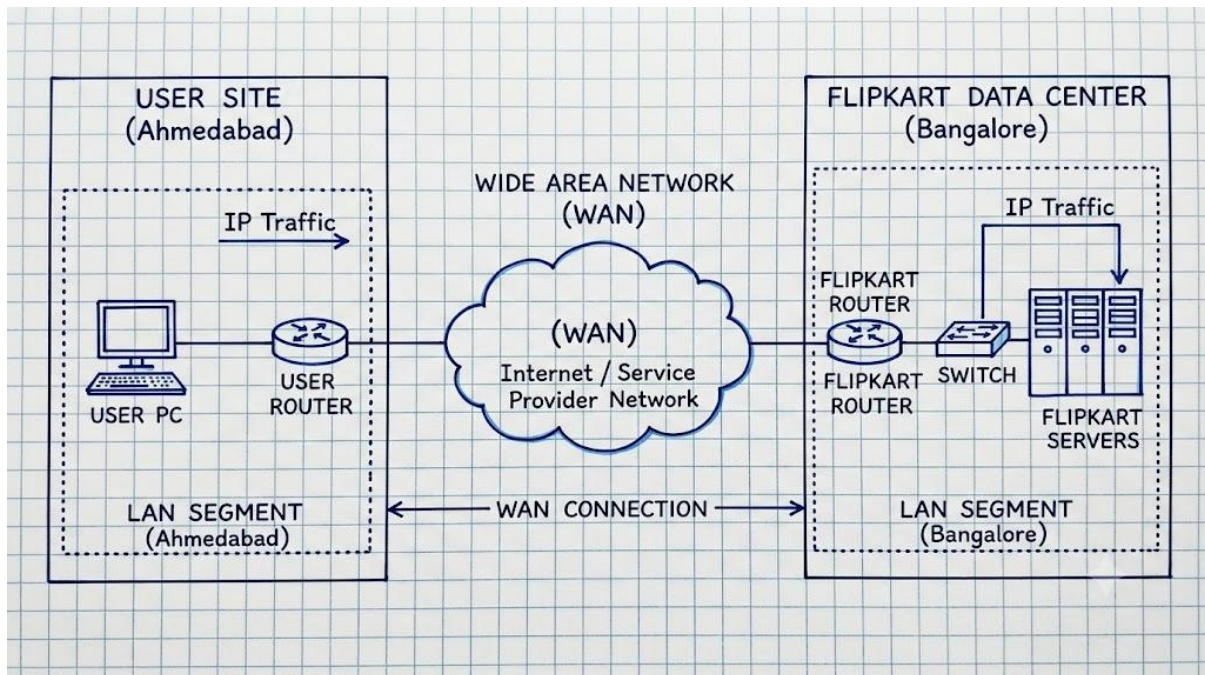
### WAN vs LAN Comparison

| Feature       | LAN (Local Area Network)                                                                           | WAN (Wide Area Network)                                                                                                       |
|---------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Coverage Area | Covers a small area such as a home, office, school, or lab.                                        | Covers large geographical areas such as cities, countries, or the whole world.                                                |
| Speed         | Usually provides higher speed because devices are nearby.                                          | Usually slower than LAN because data travels over longer distances and multiple networks.                                     |
| Ownership     | Usually owned and managed by one person or organization.                                           | Often uses infrastructure operated by telecom/Internet service providers.                                                     |
| Example       | <b>Home Wi-Fi:</b> Your phone, laptop, and smart TV connected to the same Wi-Fi router form a LAN. | <b>Instagram:</b> When you use Instagram, your phone communicates with Instagram's servers over the Internet, which is a WAN. |

### In short:

- LAN = small local network 🏠
- WAN = large network connecting distant networks 🌐

## Task-2



### Task-3

**Multiprotocol Label Switching (MPLS)** is a routing technique that directs data across a network using short, fixed-length path labels rather than complex IP destination addresses. It functions at "Layer 2.5" of the OSI model—between the data link (Layer 2) and network (Layer 3) layers—allowing it to encapsulate and forward traffic across diverse physical networks.

MPLS improves data routing compared to traditional IP routing in several key ways:

**Label-Based Switching:** Instead of requiring every intermediate router along a path to perform a full routing-table lookup based on IP destination headers, MPLS routers swap simple numeric labels in the Label Forwarding Information Base (LFIB), drastically reducing processing overhead and latency.

**Pre-Defined Paths (LSPs):** Data travels over pre-established Label Switched Paths (LSPs), avoiding unpredictable hop-by-hop decisions made by individual routers.

- **Traffic Engineering & Quality of Service (QoS):** Network administrators can prioritize latency-sensitive traffic (like VoIP, video, or live databases) and explicitly route data around congested paths, ensuring reliable performance that traditional "best-effort" IP routing cannot guarantee.

### Task-4

Based on how MPLS handles traffic flow compared to standard IP routing, here are two key routing behaviors to look for in your simulation environment:

Hop-by-Hop Route Lookups Are Replaced by Label Swapping

Standard IP Routing: Every router along the path inspects the packet's Layer 3 IP header, checks its full routing table (RIB), and calculates the next-hop IP address individually.

MPLS Behavior: Routers only inspect the outer MPLS label added by the ingress router. Intermediate routers (Provider Routers / P-Routers) consult their Label Forwarding Information Base (LFIB) to swap the existing label with a new one and forward the frame out a specific interface without performing a full IP lookup.

Traffic Follows Pre-Determined Paths (LSPs) Instead of Dynamic Hop-by-Hop Decisions

Standard IP Routing: Traffic naturally follows the shortest path calculated by an Interior Gateway Protocol (like OSPF or EIGRP). If links have equal costs, IP routing splits traffic, but it always relies on dynamic, individual router routing decisions.

MPLS Behavior: Packets follow an established Label Switched Path (LSP) setup via protocols like LDP or RSVP-TE. This allows the ingress router to direct specific traffic flows over predetermined paths, engineering traffic flow across the core network independently of traditional IGP metric calculations.

Verification Steps in Packet Tracer / GNS3:

Run a packet capture or open the Simulation Mode in Packet Tracer.

Click on a packet as it passes through a core provider router. Under the In Layer / Out Layer or PDU Details tab, check the frame header to observe the attached MPLS Label (Layer 2.5) being swapped as it transitions between interfaces.

## Task-5

### Three Benefits of MPLS for High-Traffic Companies

| Benefit                            | Explanation                                                                                                           | Real-World Scenario                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Traffic Engineering</b>      | MPLS allows companies to control the paths taken by different types of traffic and use network resources efficiently. | During a major movie release, <b>BookMyShow</b> can prioritize ticketing traffic so important requests continue flowing smoothly despite heavy traffic. |
| <b>2. Quality of Service (QoS)</b> | MPLS can prioritize delay-sensitive traffic over less important traffic, helping maintain consistent performance.     | <b>YouTube</b> can prioritize video-related traffic to reduce delays and buffering when many users are streaming simultaneously.                        |

| Benefit                                    | Explanation                                                                                                                                    | Real-World Scenario                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3. Reliable and Predictable Routing</b> | MPLS can establish predefined Label Switched Paths (LSPs), providing controlled routes and allowing faster recovery when a network path fails. | A large company's data traffic can be moved to an alternate path if the primary network link becomes unavailable, reducing service disruption. |

**In short:** MPLS helps high-traffic organizations manage **congestion, prioritize important data, and maintain reliable network performance** more effectively than basic destination-based IP routing.